Installation of Java and Execution of a Simple Hello World Program (Windows) in Oracle JDK

Software Required

- Windows Operating System
- Internet Connection
- Java Development Kit (JDK)

Step 1: Download Java JDK

1. Open a web browser.
2. Visit the official Oracle JDK website: Oracle JDK Downloads
3. Download the latest JDK for Windows (e.g., JDK 25).

Step 2: Install JDK

1. Double-click the downloaded .exe file.
2. Click **Next** and follow the installation wizard.
3. Complete the installation.

Step 3: Verify Java Installation

1. Open **Command Prompt**.
2. Type: java --version

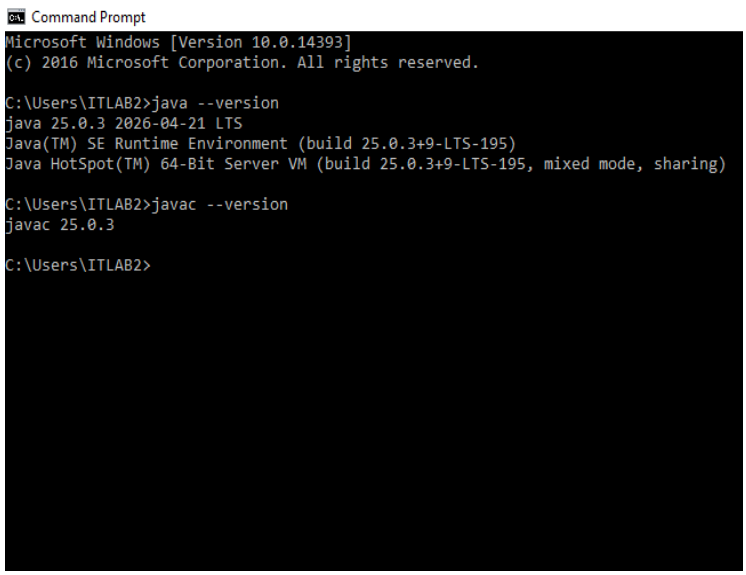
Sample Output:

java 25.0.3

3. Check the Java Compiler: javac --version

Sample Output:

javac 25.0.3



Configuring Java on Windows (Setting JAVA_HOME and PATH)

Step 1: Find Java Installation Path

After installing Java, locate the JDK folder.

Example:

C:\Program Files\Java\jdk-25.0.3

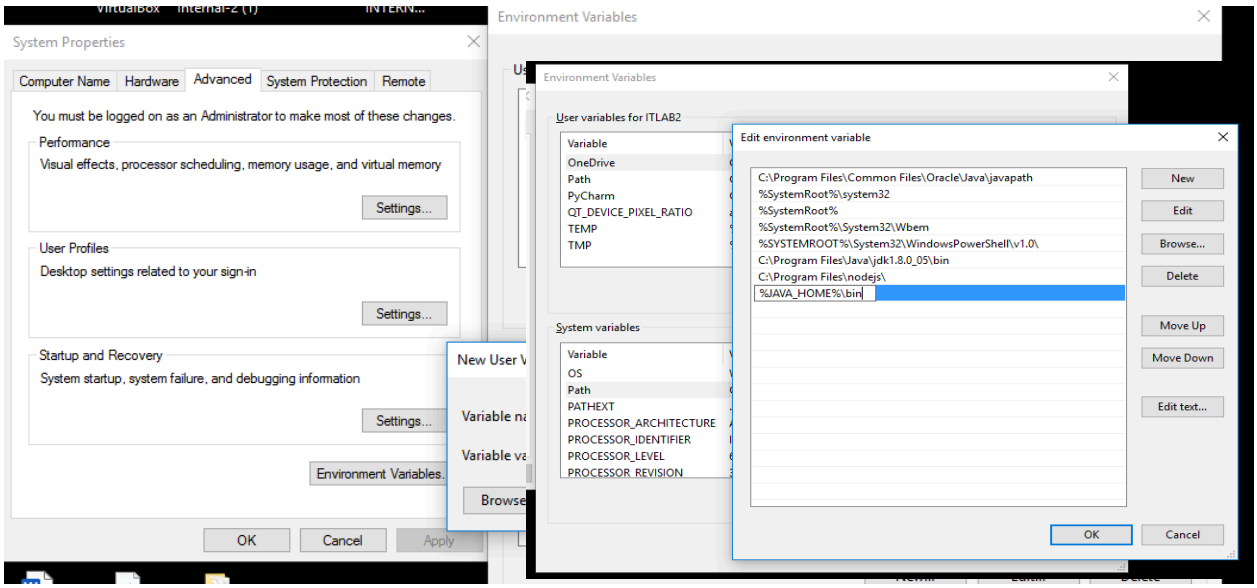
Step 2: Set JAVA_HOME

1. Press **Windows + R** → Type sysdm.cpl → Press Enter.
2. Select **Advanced** tab.

3. Click **Environment Variables**.
4. Under **System Variables**, click **New**.

Enter:

Variable Name : JAVA_HOME
Variable Value: C:\Program Files\Java\jdk-25.0.3



5. Click **OK**.

Step 3: Update PATH Variable

1. Under **System Variables**, select **Path**.
2. Click **Edit**.
3. Click **New**.
4. Add:

%JAVA_HOME%\bin

5. Click **OK → OK → OK**.

Step 4: Verify Configuration

Open a new Command Prompt and execute:

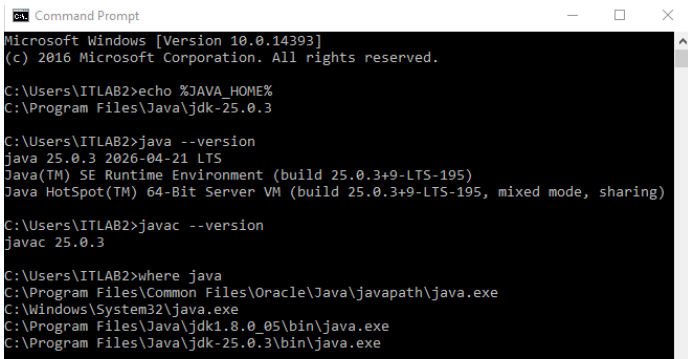
echo %JAVA_HOME%

Output:

C:\Program Files\Java\jdk-25.0.3

Check Java version:

java --version



Check Java compiler:

javac --version

Sample Output:

java 25.0.3
javac 25.0.3

Execution of ‘Hello’ program in Java:

1. Create a Java file:

Create a file named Hello.java:

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello");
    }
}
```

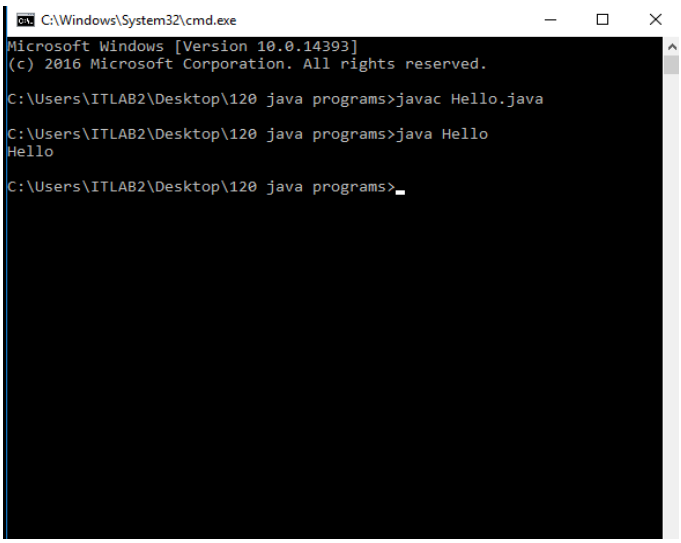
Store it in the format of .java

2. Open Command Prompt:

Type javac Hello.java

It would create a ‘.class’ file for the program.

3. Run the program:
Type java Hello
Sample Output:
Hello



Installation of Java and Execution of a Simple Hello World Program (Windows) in OpenJDK

Software Required:

- Windows Operating System
- Internet Connection
- OpenJDK
- Command Prompt

Step 1: Download OpenJDK

1. Open any web browser.
2. Visit the official **OpenJDK** website or **Eclipse Adoptium** website.
3. Download the latest stable **OpenJDK for Windows (64-bit)**.

Step 2: Install OpenJDK

1. Double-click the downloaded installer.
2. Follow the installation wizard.
3. Choose the installation location if required.
4. Click **Install** and wait until the installation is completed.
5. Click **Finish**.

Step 3: Configure Java Environment Variables

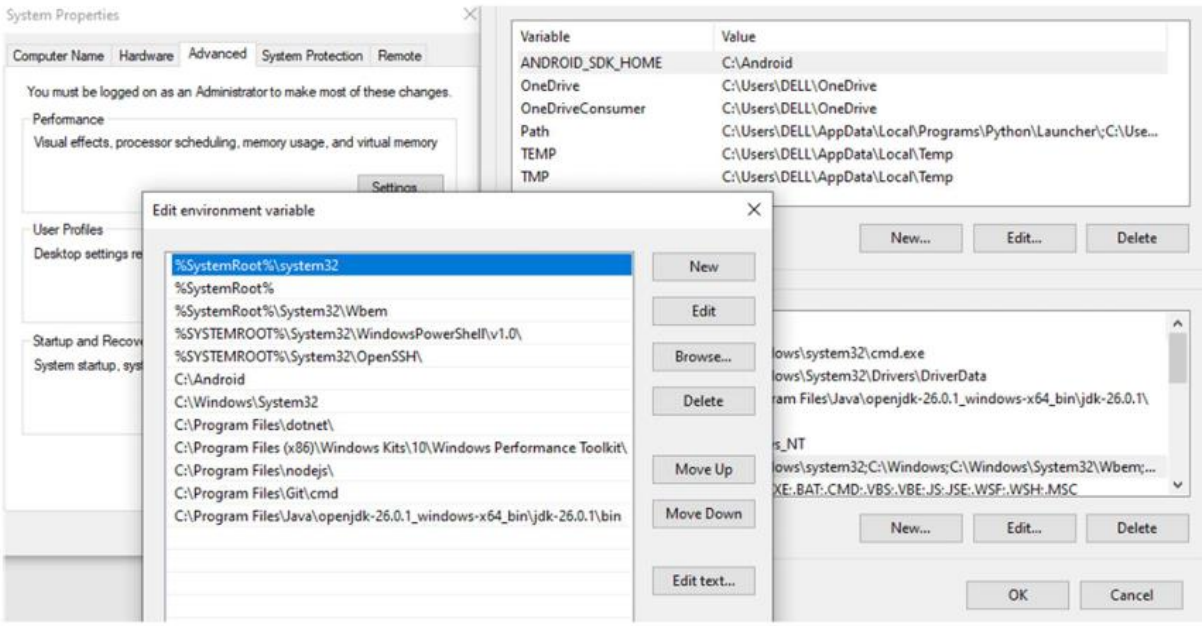
Set JAVA_HOME

1. Press **Windows + R**, type **sysdm.cpl**, and press **Enter**.
2. Open the **Advanced** tab.
3. Click **Environment Variables**.
4. Under **System Variables**, click **New**.
5. Enter the following:

Variable Name: JAVA_HOME

Variable Value: *(Example)*

C:\Program Files\Eclipse Adoptium\jdk-21



(Use the path where OpenJDK is installed on your system.)

Update the PATH Variable

1. Under **System Variables**, select **Path**.
2. Click **Edit**.
3. Click **New**.
4. Add:

%JAVA_HOME%\bin

5. Click **OK** to save all the changes.

Step 4: Verify the Installation

1. Open **Command Prompt**.
2. Type:

java --version

Sample Output

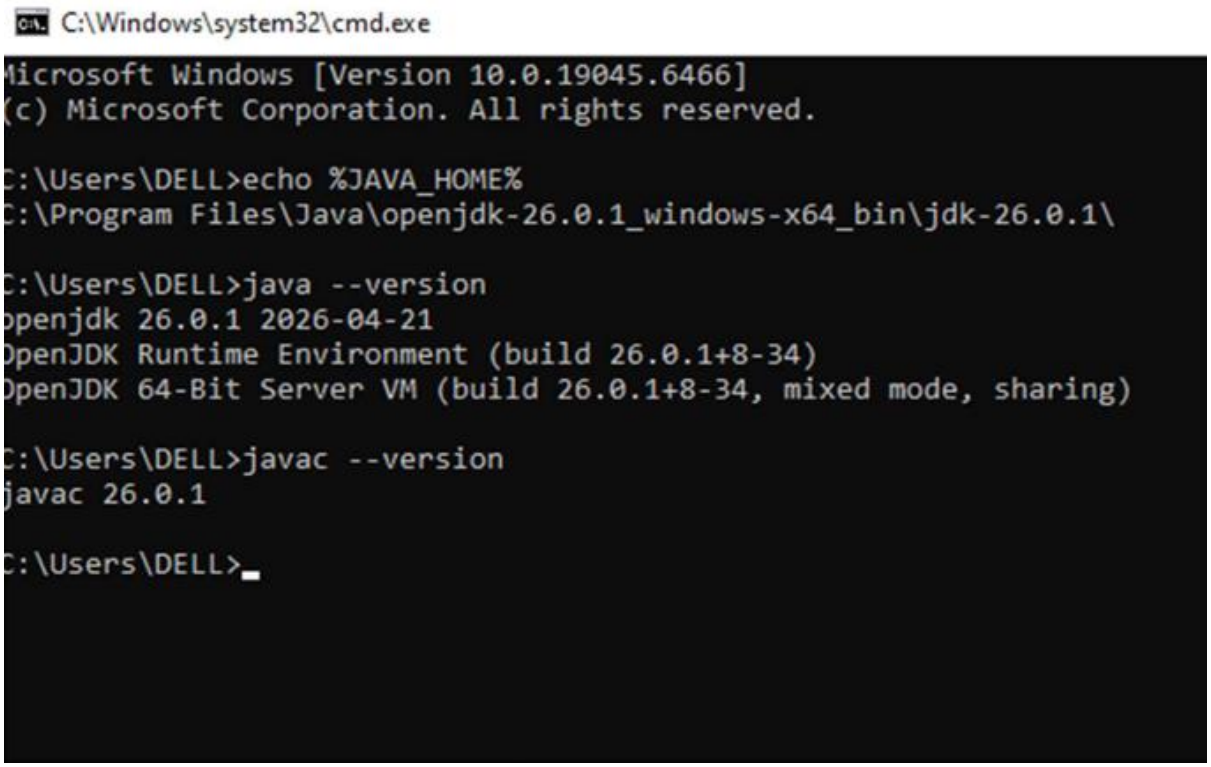
openjdk 21.0.11

3. Now check the Java compiler.

javac --version

Sample Output

javac 21.0.11



Execution of a Simple Hello World Program

Step 1: Create a Java Program

Create a file named **Hello.java** and type the following program:

```
public class Hello {
   public static void main(String[] args) {
      System.out.println("Hello World");
   }
}
```

Save the file with the **.java** extension.

Step 2: Compile the Program

Open Command Prompt and navigate to the folder where the file is saved.

Compile the program using:

javac Hello.java

A file named **Hello.class** will be generated after successful compilation.

Step 3: Run the Program

Execute the program using:

java Hello

Output

Hello World

